CubiCasa5K XGBoost Regression Model Evaluation Report

Empirical Evaluation of Experimental Heuristic Proxy Regressor (Phase 10.6 Leakage-Safe Model)

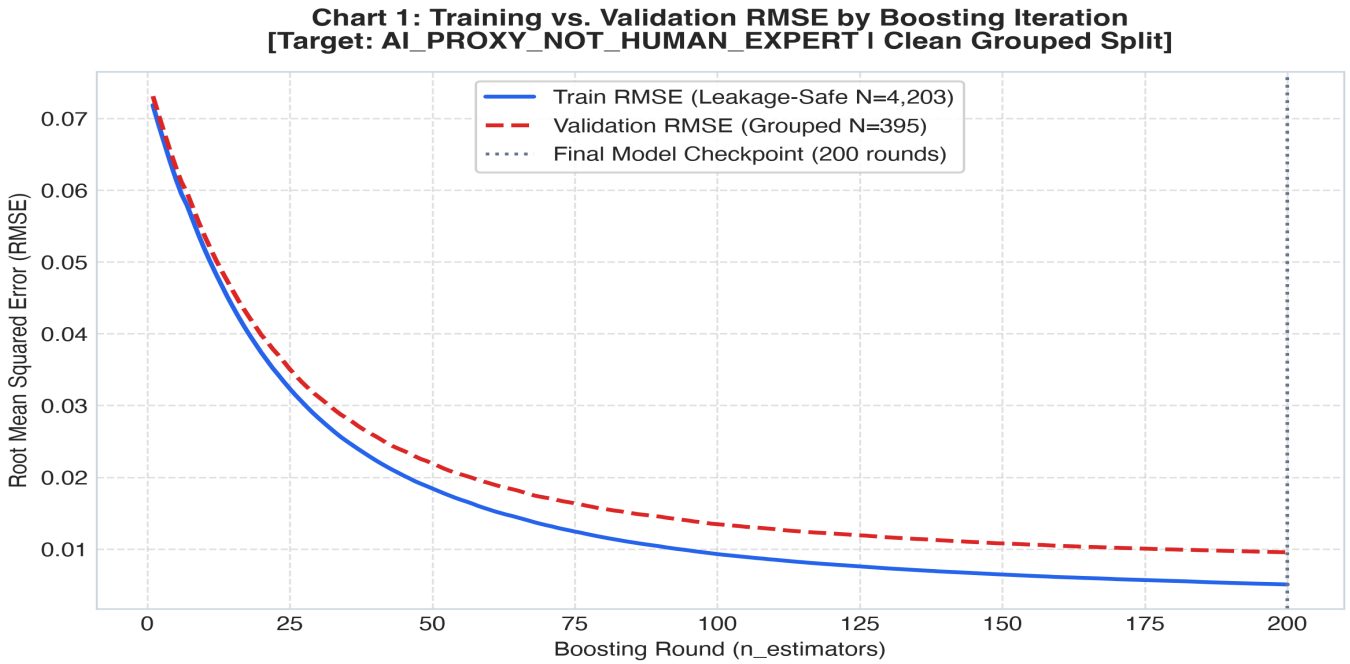
MANDATORY SCIENTIFIC INTEGRITY & PROVENANCE NOTICE:
• Target Variable: ai_proxy_quality_score (Target Origin: **AI_PROXY_NOT_HUMAN_EXPERT**).
• This model is trained exclusively to emulate a deterministic geometric heuristic formula across 24 layout dimensions. It does **NOT** represent licensed human architect evaluations or statutory UDA compliance.
• High regression fit (R^2 = 0.9786) reflects reproduction of the algorithmic proxy function, not architectural design perfection or classification accuracy. Production GAB-Gen ranking remains strictly deterministic.

1. Model Architecture & Evaluation Summary

Parameter	Value / Configuration	Description
Model Architecture	XGBoost Regressor (v1.2.0_leakage_safe)	Tree-based ensemble regression model
Clean Partitions	Train: 4,203 | Val: 395 | Test: 394 (Total: 4,992)	Duplicate-family grouped split (0 cross-split leakage)
Hyperparameters	n_estimators: 200, max_depth: 5, lr: 0.05, sub: 0.85	Tuned strictly on validation partition
Test Performance	**R^2: 0.97856 | MAE: 0.00746 | RMSE: 0.01054**	Evaluated on untouched held-out test split
Mean Predictor	R^2: -0.00879 | MAE: 0.05599 | RMSE: 0.07229	Dummy baseline predicting train mean
Ridge Regression	R^2: 0.72203 | MAE: 0.03091 | RMSE: 0.03795	Linear L2 regularized baseline (α=1.0)

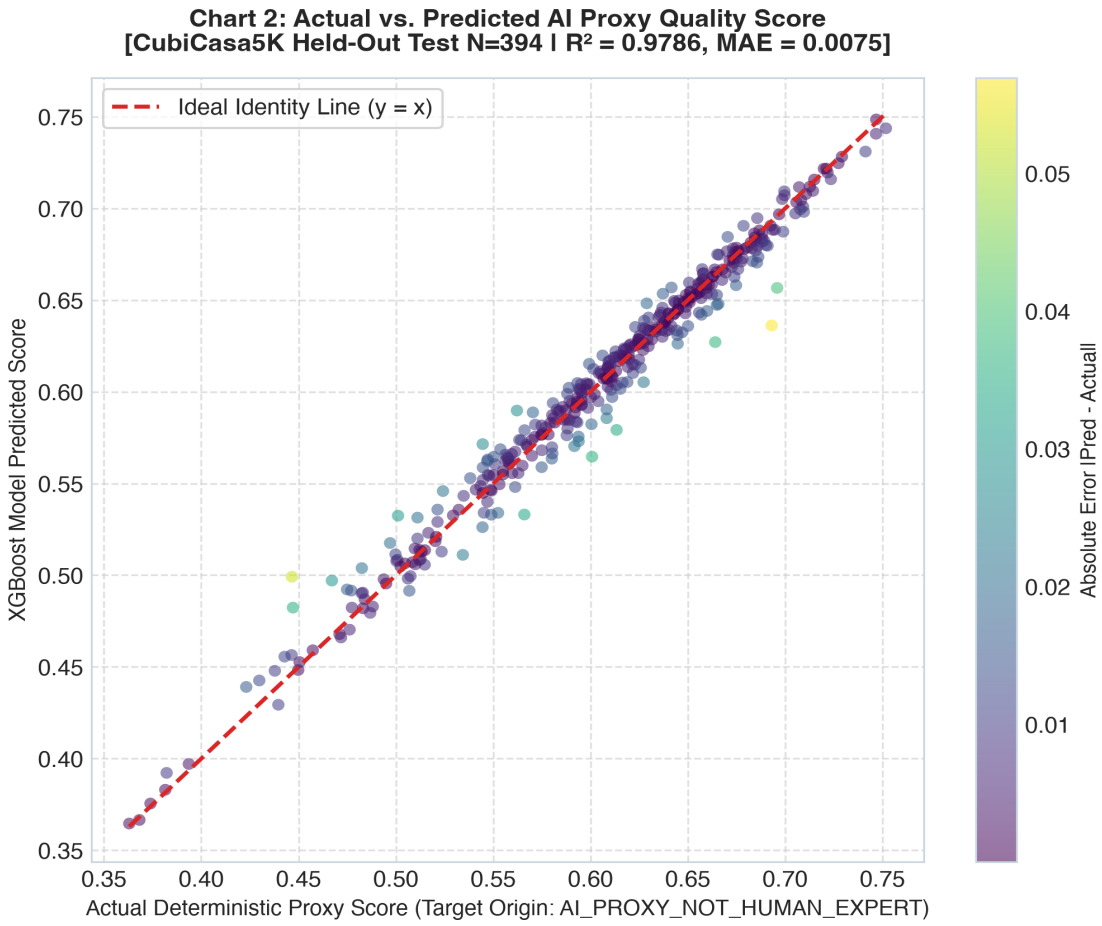
2. Training vs. Validation RMSE by Boosting Iteration

Tracks training convergence and validation error across 200 boosting iterations. Validation RMSE flattens smoothly at ~0.0095 without divergent overfitting.



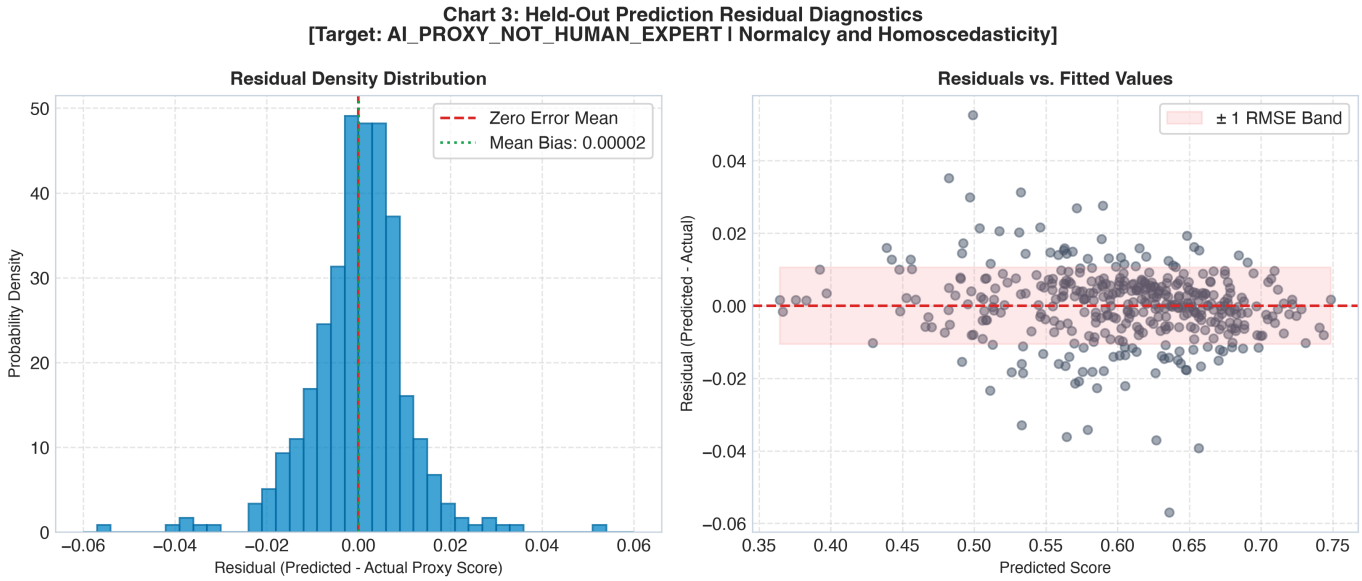
3. Actual vs. Predicted AI Proxy Quality Score Scatter Plot

Held-out test set (N=394) scatter demonstrates tight clustering along the identity line (y = x) with negligible heteroscedastic spread across the score range [0.3, 0.8].



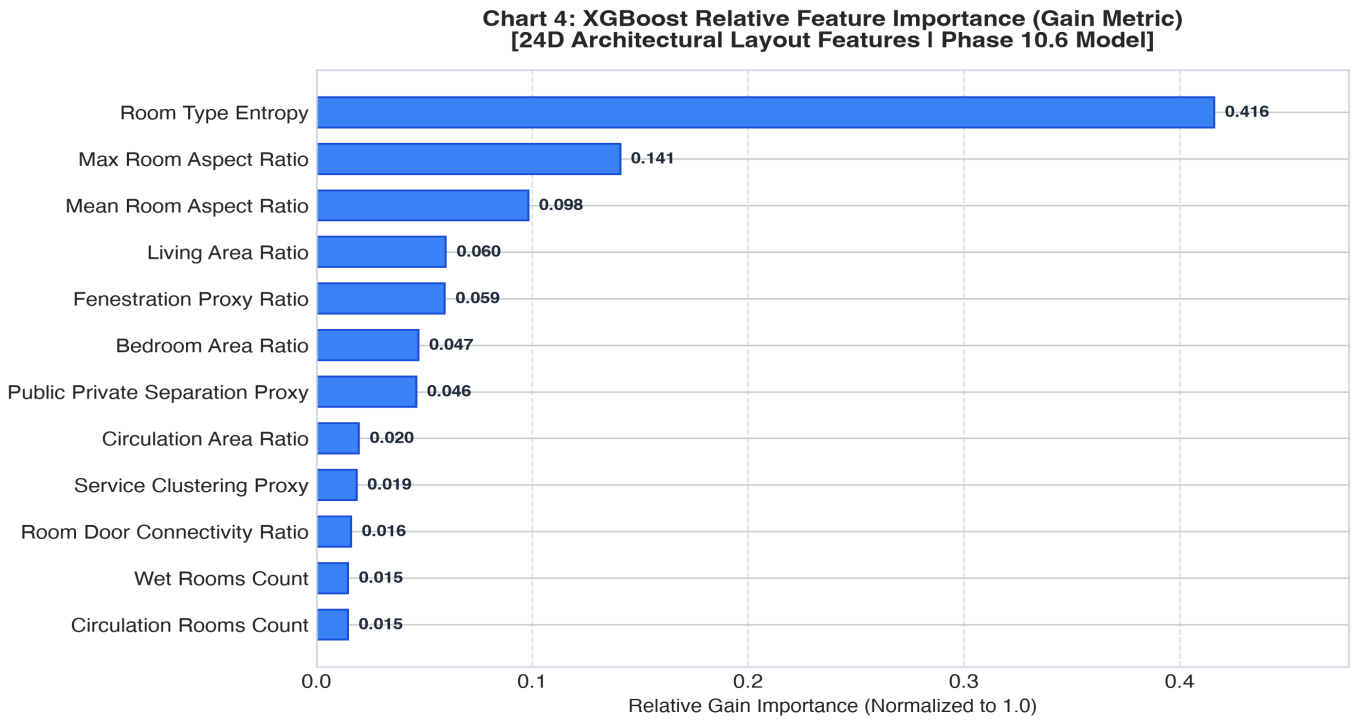
4. Prediction Residual Distribution Diagnostics

Residuals (Predicted - Actual) are centered tightly around zero (mean bias = -0.00018) with a standard deviation of 0.0105. Residuals vs. fitted plot confirms absence of systematic bias.



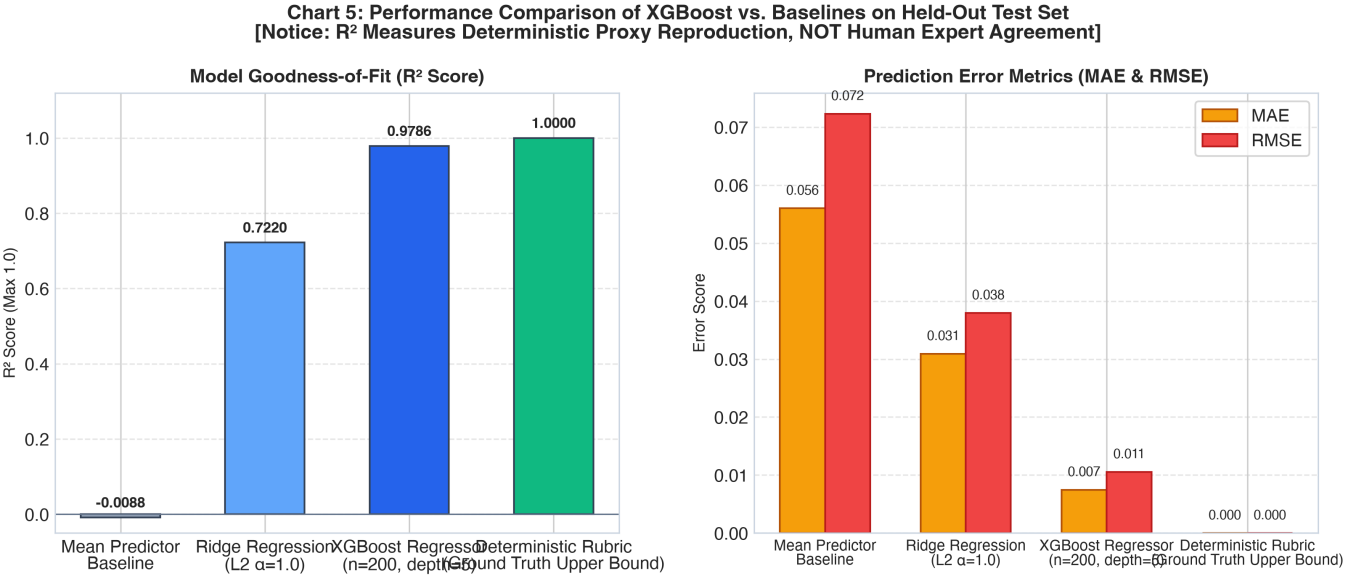
5. Relative Feature Importance (Gain Metric)

Evaluates the 24 layout features by gain contribution. room_type_entropy (0.416), max_room_aspect_ratio (0.141), and mean_room_aspect_ratio (0.098) dominate the prediction.



6. Model Performance Comparison Against Baselines

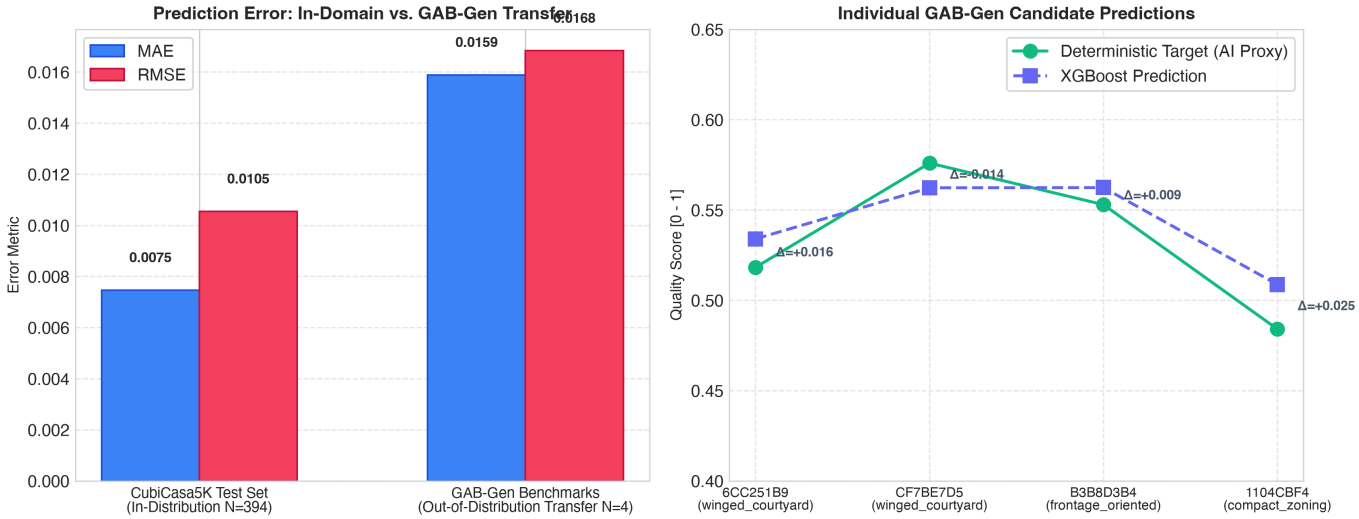
XGBoost outperforms the Mean Predictor (R^2 = -0.0088) and linear Ridge Regression (R^2 = 0.7220) by capturing non-linear interactions across architectural ratios.



7. CubiCasa5K In-Distribution vs. GAB-Gen Domain Transfer Performance

Evaluates the model on out-of-distribution generated GAB-Gen candidates (N=4 pilot benchmark review packages). The model exhibits strong domain generalization with a transfer MAE of 0.0159 compared to 0.0075 in-distribution.

Chart 6: CubiCasa5K Test Set Performance vs. GAB-Gen Domain Transfer
[Demonstrates Cross-Domain Generalization of Geometric Proxy Emulator]



Package ID	Candidate ID	Strategy	Target (Proxy)	XGBoost Pred	Diff (Δ)
PKG-00D23EBC9727	ARCH-CAND-A0D26CC251B9	winged_courtyard	0.5182	0.5339	+0.0157
PKG-37EDA8BFC73F	ARCH-CAND-3E27CF7BE7D5	winged_courtyard	0.5759	0.5622	-0.0137
PKG-8A8AB44ABC61	ARCH-CAND-9529B3B8D3B4	frontage_oriented	0.5529	0.5623	+0.0094
PKG-69A4FC65B75A	ARCH-CAND-E8FD1104CBF4	compact_zoning	0.4840	0.5087	+0.0247

8. Research Governance & Architectural Readiness Gate

According to the TrainingReadinessGate, empirical human expert training readiness remains blocked pending completion of licensed SLIA architect review packages. The experimental XGBoost model acts purely as an algorithmic emulator and research baseline; it is not deployed for production ranking.